

A Framework for Two-Stage Enrichment Trial Designs

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Abstract

In this paper, we develop a framework for a two-stage enrichment design to investigate a treatment effect in a clinical trial. A learning stage identifies a potentially benefitting subgroup, and that information is carried forward to a confirmatory stage with adapted enrollment criteria. Final inference combines data from both stages, and subgroup performance is evaluated using false inclusion and false exclusion rates. In stage 1, to estimate a benefitting subgroup, we propose a non-parametric regression level-set approach that relaxes monotonicity assumptions among continuous biomarkers and the treatment effect. The resulting optimal subgroup is designed to balance statistical validity with ethical considerations in patient enrollment. We describe potential methods to conduct the second stage and relevant inference, topics that should be considered research in progress.

Keywords: Level-set Estimation, Nonparametric Regression, Two-stage Inference, Subgroup Selection

1 Introduction

Precision medicine has increasingly reshaped the design and analysis of clinical trials by recognizing that treatment effects may vary substantially across patient subpopulations. Traditional randomized clinical trials typically focus on estimating an average treatment effect in the overall population, which can mask clinically meaningful benefits experienced by only a subset of patients. As a result, potentially effective therapies may fail to demonstrate efficacy when evaluated at the population level, despite benefiting patients with specific biomarker profiles. This challenge has motivated the development of enrichment designs, which aim to improve trial efficiency by restricting or adapting enrollment to subpopulations that are more likely to benefit from treatment.

Adaptive enrichment designs (Simon and Simon (2013, 2017)) extend this idea by allowing subgroup definitions to be updated using interim data. In a typical two-stage design, patients are initially enrolled from the full population in a learning stage, during which evidence for a treatment-sensitive subgroup is assessed. If such a subgroup is identified, enrollment criteria are modified in a confirmatory stage to only include patients from the estimated benefitting subpopulation. By adapting eligibility criteria rather than randomization, adaptive enrichment designs preserve the integrity of randomized comparisons while improving both statistical efficiency and ethical allocation of patients to treatment.

Despite their appeal, adaptive enrichment designs present substantial statistical challenges. A central difficulty lies in identifying benefitting subgroups when treatment effects depend on multiple continuous biomarkers through complex, nonlinear relationships. Many existing approaches rely on parametric models or monotonicity assumptions linking biomarkers to treatment response, which may be unrealistic in practice and can lead to biased subgroup identification. Moreover, data-driven subgroup selection introduces the risk of inflated error rates when the same data are used for both subgroup identification and confirmatory inference.

To address these challenges, this article proposes a nonparametric framework for adaptive enrichment designs based on regression level-set estimation (Reeve et al. (2023), Armstrong and Shen (2023)). The benefitting subgroup is defined as the region of the biomarker space where the conditional treatment effect exceeds a clinically meaningful threshold. Treatment effects are estimated using kernel-based local linear regression, allowing flexible modeling of heterogeneous and non-monotone biomarker–treatment relationships. Rather than relying on undersmoothing to control bias, we explicitly estimate and correct the smoothing bias, enabling stable estimation of treatment-effect level sets while maintaining reasonable variance properties.

We construct a one-sided confidence set for the treatment-effect level set that controls false exclusion of truly benefitting patients with high probability. This approach yields a principled method for subgroup identification in the learning stage and provides a transparent mechanism for adapting enrollment in the confirmatory stage. Subgroup performance is evaluated using false inclusion and false exclusion rates, reflecting both statistical validity and ethical considerations in patient selection. The proposed methodology offers a flexible and interpretable framework for adaptive

enrichment designs with continuous biomarkers, bridging nonparametric estimation, level-set inference, and confirmatory analysis in clinical trials.

2 Problem Setup and Framework

We consider a two-stage adaptive enrichment design for a randomized clinical trial with baseline biomarkers. The goal is to identify a subgroup of patients who are likely to benefit from treatment and to confirm the treatment effect within this subgroup while maintaining valid statistical inference.

2.1 Two-Stage Enrichment Design

The trial proceeds in two stages: a learning stage and a confirmatory stage. In the learning stage, patients are enrolled from the full target population and randomized to either treatment or control. Data collected at this stage are used to explore heterogeneity in treatment response and to assess whether a benefitting subgroup exists.

If evidence for a benefitting subgroup is found, the trial enters a confirmatory stage in which eligibility criteria are adapted to preferentially enroll patients whose biomarker profiles are consistent with the estimated subgroup. Importantly, the randomization mechanism itself is not modified; treatment assignment remains randomized within each stage. This separation between adaptation of eligibility and preservation of randomization ensures that causal interpretation of treatment effects is retained.

The primary estimand of interest is the conditional treatment effect as a function of predicting biomarkers. Let $Y \in \mathbb{R}$ denote the clinical outcome, $T \in \{0, 1\}$ the treatment indicator, and $\mathbf{x} \in \mathbb{R}^d$ a vector of realized biomarkers. Let $\mu_T(\mathbf{x}) = \mathbb{E}(Y \mid \mathbf{X} = \mathbf{x}, T = t), t = 0, 1$. The conditional treatment effect can then be defined as

$$\eta(\mathbf{x}) = \mu_1(\mathbf{x}) - \mu_0(\mathbf{x}), \quad (1)$$

which characterizes how treatment benefit varies across the biomarker space between two groups: treatment vs. placebo. The relative risk or odds ratio could also be used.

2.2 Subgroup Definition via Level Sets

We define the benefitting subgroup L_τ through a level-set representation of the $\eta(\mathbf{x})$. For a given threshold τ representing a clinically or scientifically meaningful treatment effect, the target subgroup is

$$L_\tau = \{\mathbf{x} \in \mathcal{X} : \eta(\mathbf{x}) \geq \tau\}, \quad (2)$$

where $\mathcal{X} = \text{supp}(\mathbf{X}) \subset \mathbb{R}^d$. This formulation allows subgroup membership to depend on multiple biomarkers through potentially nonlinear and non-monotone relationships.

A nonparametric approach to subgroup selection is the optimal subgroup selection (OSS) framework of [Reeve et al. \(2023\)](#), which defines the benefitting subgroup in (2) under a conservative containment constraint to control false exclusion. The

method avoids parametric assumptions and provides rigorous error control, but it is primarily developed for a single, fixed subgroup identification step rather than adaptive enrichment across stages. As a result, its partition-based construction may yield fragmented subgroups in finite samples or higher-dimensional biomarker spaces, motivating complementary kernel-based regression level-set approaches that naturally support adaptive subgroup refinement.

We propose an estimator $\hat{L}_\tau$ of the level set L_τ for the learning stage and use the estimation to guide enrollment in the confirmatory stage. Two types of errors naturally arise in this context. False exclusion occurs when patients who truly belong to L_τ are omitted from $\hat{L}_\tau$, potentially denying beneficial treatment. False inclusion occurs when patients outside L_τ are included in $\hat{L}_\tau$, which may reduce efficiency but can remain ethically acceptable when treatment is not harmful. The methodology developed in this paper prioritizes control of false exclusion while limiting false inclusion, reflecting both statistical and ethical considerations inherent in adaptive enrichment designs.

Relative to partition-based optimal subgroup selection methods (Reeve et al. (2023)), the proposed kernel-based estimator produces smoother and more connected subgroup regions. This structural stability is particularly important in moderately sized clinical trials, where highly restrictive subgroup definitions may complicate patient enrollment and practical trial conduct. In comparison with undersmoothing strategies commonly used in nonparametric inference Armstrong and Shen (2023), explicit bias correction avoids unnecessary variance inflation and leads to more stable level-set boundaries, especially near clinically relevant thresholds.

2.3 Threshold Estimation

A central component of regression level-set based enrichment is the specification of the threshold that determines which covariate profiles are considered to exhibit a clinically relevant treatment effect.

(i) Fixed clinically meaningful threshold.

The simplest specification treats τ as a fixed constant determined externally based on clinical considerations. In this case,

$$\tau = c,$$

where c represents a minimum treatment effect deemed clinically meaningful. This definition is attractive for its interpretability and regulatory transparency, but it requires the effect threshold to be specified *a priori* and does not adapt to the distribution of treatment effects in the enrolled population.

(ii) Average-effect calibrated threshold.

An alternative approach defines the threshold implicitly through a population-level performance criterion. Given a target effect size c , the threshold is defined as

$$\tau^* = \inf \{s : \mathbb{E}[Y \mid \eta(\mathbf{X}) \geq s] \geq c\}.$$

This definition ensures that the *average* treatment effect among patients included in the subgroup meets the clinical target, even if some individuals exhibit weaker effects. This idea appears in a parametric enrichment frameworks, such as those studied by [Lin et al. \(2021\)](#). The resulting subgroup is data-adaptive and directly linked to the expected benefit among enrolled patients.

(iii) Utility-based threshold.

A third definition treats the threshold as the solution to an explicit optimization problem that balances subgroup prevalence and treatment magnitude. Let

$$U(L_\tau) = [\mathbb{P}(\mathbf{X} \in L_\tau)]^\gamma \mathbb{E}[Y \mid \mathbf{X} \in L_\tau], \quad \gamma \in [0, 1]$$

denote a utility function that increases with both subgroup size and average treatment effect. The optimal threshold is defined as

$$\tilde{\tau} = \arg \max_s U(L_s).$$

This formulation, studied in adaptive enrichment designs by [Joshi et al. \(2020\)](#), [Lai et al. \(2019\)](#), yields a data-driven subgroup that explicitly trades off effect strength against enrollment efficiency. Unlike fixed-threshold approaches, the utility-based definition is design-oriented and aligns subgroup selection with operational objectives of the trial.

Other threshold definition can be found in [Stallard \(2023\)](#). These threshold specifications are not mutually exclusive. Fixed thresholds emphasize clinical interpretability, average-effect thresholds target population-level benefit, and utility-based thresholds prioritize efficiency. The regression level-set framework adopted here accommodates all three choices, allowing the threshold to be specified externally or estimated from data without altering the underlying inferential machinery.

2.4 Enrollment Adaptation

Let $\hat{L}_\tau$ denote the estimated benefitting subgroup obtained from the learning stage. In the confirmatory stage, patient enrollment is adapted by restricting eligibility to individuals whose baseline biomarker profiles satisfy $\mathbf{X} \in \hat{L}_\tau$. This adaptation aims to concentrate enrollment on patients with a higher likelihood of treatment benefit, thereby improving efficiency and ethical allocation of participants.

From an ethical perspective, the use of a one-sided confidence set for subgroup identification prioritizes protection against false exclusion, reducing the risk of denying potentially beneficial treatment to eligible patients.

2.5 Treatment Effect Estimation

After enrollment is restricted to the estimated benefitting subgroup $\hat{L}_\tau$, the confirmatory stage focuses on inference for the treatment effect within this subpopulation. Let D_2 denote data accrued in the confirmatory stage, and let D_1 denote the learning-stage

data used to estimate $\widehat{L}_\tau$. The primary inferential goal is to test the null hypothesis of no treatment effect among patients satisfying $\mathbf{X} \in \widehat{L}_\tau$.

3 Nonparametric Regression Level-Set Methodology

This section introduces the proposed nonparametric framework for estimating treatment-effect level sets in the learning stage of a two-stage adaptive enrichment design. The methodology combines a kernel-based regression technique, explicit bias correction, and uniform error control to construct one-sided confidence sets for benefitting subgroups.

3.1 Treatment Effect Estimation

Let $(Y_i, \mathbf{X}_i, T_i), i = 1, \dots, n$, denote independent observations from the learning stage, where $Y_i \in \mathbb{R}$ is the outcome, $\mathbf{X}_i \in \mathbb{R}^d$ is a vector of biomarkers, and $T_i \in \{0, 1\}$ indicates treatment assignment.

We estimate $\mu_t(\mathbf{x})$ defined in (1) using kernel-based local linear regression. Specifically, for a given evaluation point $\mathbf{x} \in \mathbb{R}^d$, the local linear estimator $\widehat{\mu}_t(\mathbf{x})$ is obtained by solving

$$\left(\widehat{a}_t(\mathbf{x}), \widehat{b}_t(\mathbf{x})\right) = \arg \min_{a \in \mathbb{R}, b \in \mathbb{R}^d} \sum_{i: T_i=t} K_h(\mathbf{X}_i - \mathbf{x}) \{Y_i - a - b^\top (\mathbf{X}_i - \mathbf{x})\}^2, \quad (3)$$

where $K_h(\mathbf{u}) = h^{-d}K(\mathbf{u}/h)$, $K(\cdot)$ is a symmetric kernel function such that $\int K(\mathbf{u})d\mathbf{u} = 1$. We adopt a non-compact support kernel satisfying $K(\mathbf{u}) > 0$ for all $\mathbf{u} \in \mathbb{R}^d$. Assume that K is symmetric, three times continuously differentiable, and has finite second moments so that $M_2(K) = \int \mathbf{u}\mathbf{u}^\top K(\mathbf{u})d\mathbf{u}$ is well defined. We employ the Gaussian kernel in the simulation study, allowing information from all enrolled patients to contribute to the estimation procedure. Let $h > 0$ denote the bandwidth parameter, and in the multi-dimensional biomarker setting, we assign the same bandwidth across different biomarkers. We let the estimate of the regression function at $\mathbf{x}$: $\widehat{\mu}_t(\mathbf{x}) = \widehat{a}_t(\mathbf{x})$ be the local linear estimator that solves the weighted least-squares problem in (3) (Fan and Yao (1998)). So, the conditional treatment effect function is then estimated as

$$\widehat{\eta}(\mathbf{x}) = \widehat{\mu}_1(\mathbf{x}) - \widehat{\mu}_0(\mathbf{x}).$$

Local linear regression provides automatic boundary correction and reduced bias relative to local constant estimators, making it well suited for estimating smooth but potentially complex treatment-effect surfaces.

3.2 Bias Correction Strategy

Kernel-based regression estimators are inherently biased due to smoothing. For the treatment-effect estimator $\hat{\eta}(\mathbf{x})$, the estimation error can be decomposed as

$$\hat{\eta}(\mathbf{x}) - \eta(\mathbf{x}) = \{\hat{\eta}(\mathbf{x}) - \mathbb{E}[\hat{\eta}(\mathbf{x})]\} + \{\mathbb{E}[\hat{\eta}(\mathbf{x})] - \eta(\mathbf{x})\},$$

where the first term represents stochastic variation and the second term corresponds to bias. A common approach to mitigating bias in nonparametric inference is under-smoothing as implemented in [Armstrong and Shen \(2023\)](#), which forces the bandwidth to shrink faster than the optimal rate so that the bias becomes asymptotically negligible. However, under-smoothing inflates variance of the estimator, destabilizes boundary estimation for level sets, and can substantially reduce power in finite samples. These drawbacks are particularly problematic in clinical trial settings with moderate sample sizes.

Instead, [Qiao and Polonik \(2019\)](#) considered a bias correction mechanism ([Hall \(1992\)](#)) for a density function $\hat{f}$ by setting the de-biased estimator $\hat{f}_{bc} = \hat{f} - \hat{\beta}$, and we extend this strategy to the conditional expectation function: (e.g. $\mathbb{E}(Y | X, T)$). The leading bias term of local linear estimator $\hat{\eta}(\mathbf{x})$ is of order h^2 . Assuming $\eta(x)$ is three times continuously differentiable on $\mathcal{X}$, a second-order Taylor expansion yields the following bias representation:

$$\begin{aligned} \mathbb{E}[\hat{\eta}(\mathbf{x})] - \eta(\mathbf{x}) &= \beta(\mathbf{x}) + o(h^2), \\ \beta(\mathbf{x}) &= \frac{h^2}{2} \text{tr}(\nabla^2 \eta(\mathbf{x}) M_2(K)), \end{aligned}$$

Accordingly, we estimate the leading bias term $\beta(\mathbf{x})$ using the plug-in estimator

$$\hat{\beta}(\mathbf{x}) = \frac{h^2}{2} \text{tr}(\widehat{\nabla^2 \eta}(\mathbf{x}) M_2(K)),$$

where $\widehat{\nabla^2 \eta}(\mathbf{x})$ is obtained from a local quadratic regression fit. In the one-dimensional simulation setting considered in [Section 5](#), this reduces to

$$\hat{\beta}(x) = \frac{h^2}{2} \mu_2(K) \hat{\eta}''(x) = \frac{h^2}{2} \mu_2(K) \hat{\mu}_1''(x) - \frac{h^2}{2} \mu_2(K) \hat{\mu}_0''(x)$$

with $\mu_2(K) = \int u^2 K(u) du$ denotes the second moment of the kernel function. We estimate $\hat{\mu}_t''(x)$ by fitting a local quadratic regression ([Fan and Gijbels \(1996\)](#)) within the treatment group $t = \{0, 1\}$. Briefly, for each evaluation point x , solve

$$(\hat{a}_t(x), \hat{b}_t(x), \hat{c}_t(x)) = \arg \min_{a,b,c} \sum_{i: T_i=t} K_h(X_i - x) \left\{ Y_i - a - b(X_i - x) - c(X_i - x)^2 \right\}^2,$$

the estimated second derivative is $\hat{\mu}_t''(x) = 2\hat{c}_t(x)$. So the plug-in bias estimator is

$$\hat{\beta}(x) = h^2 \mu_2(K) \hat{c}_1(x) - h^2 \mu_2(K) \hat{c}_0(x).$$

Hence, we construct the bias-corrected estimator $\hat{\eta}^*(\mathbf{x})$ as

$$\hat{\eta}^*(\mathbf{x}) = \hat{\eta}(\mathbf{x}) - \hat{\beta}(\mathbf{x}), \quad \text{and} \quad \hat{\eta}^*(\mathbf{x}) - \eta(\mathbf{x}) = \hat{\eta}(\mathbf{x}) - \mathbb{E}(\hat{\eta}(\mathbf{x})) + o_p(h^2)$$

This correction allows the use of bandwidths chosen for optimal mean squared error while maintaining valid inference for level-set estimation.

3.3 One-Sided Confidence Sets

To estimate the standard deviation of the bias-corrected local linear estimator $\hat{\eta}^*(\mathbf{x})$, define the residuals $\hat{e}_{it} = Y_i - \hat{\mu}_t(\mathbf{X}_i)$, $t \in \{0, 1\}$, where $\hat{e}_{it}$ denotes the residual for observation i in treatment group t . Let

$$s_t^2(\mathbf{x}) = \text{Var}(Y \mid \mathbf{X} = \mathbf{x}, T = t), \quad t \in \{0, 1\},$$

denote the conditional local residual variance within treatment group t . We estimate this quantity using the kernel-smoothed residual variance estimator

$$\hat{s}_t^2(\mathbf{x}) = \frac{\sum_{1 \leq i \leq n, T_i = t} K_h(\mathbf{X}_i - \mathbf{x}) \hat{e}_{it}^2}{\sum_{1 \leq i \leq n, T_i = t} K_h(\mathbf{X}_i - \mathbf{x})}, \quad t \in \{0, 1\}.$$

with $K_h(\mathbf{u}) = h^{-d} K(\mathbf{u}/h)$.

Under the standard asymptotic variance approximation for local linear regression estimators (e.g., [Härdle \(1990\)](#); [Fan and Gijbels \(1996\)](#)), and since the simulation setting assumes $d = 1$ and $X \sim \text{Uniform}(0, 1)$ so that $f_X(x) \equiv 1$ on $\mathcal{X}$, the standard deviation of $\hat{\eta}^*(x)$ takes the form:

$$\sigma(x) = \left[R(K) \left\{ \frac{s_1^2(x)}{nh} + \frac{s_0^2(x)}{nh} \right\} \right]^{1/2},$$

We therefore define the plug-in standard deviation estimator

$$\hat{\sigma}(x) = \left[R(K) \left\{ \frac{\hat{s}_1^2(x)}{nh} + \frac{\hat{s}_0^2(x)}{nh} \right\} \right]^{1/2}.$$

Although the simulation study is conducted with finite samples, we employ the standard asymptotic variance approximation from local linear regression theory to define the empirical standard deviation estimator. Using this estimator, we construct the corresponding studentized error process

$$\hat{Z}_n(\mathbf{x}) = \frac{\eta(\mathbf{x}) - \hat{\eta}^*(\mathbf{x})}{\hat{\sigma}(\mathbf{x})}, \quad \mathbf{x} \in \mathcal{X}. \quad (4)$$

The key inferential ingredient of our approach is uniform control over the supremum of this process. Let $C_{1-\alpha}$ be a critical value satisfying

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{X}} \widehat{Z}_n(\mathbf{x}) \leq C_{1-\alpha} \right) \geq 1 - \alpha. \quad (5)$$

Such a critical value can be approximated using a Gaussian multiplier bootstrap ([Armstrong and Shen \(2023\)](#), [Chernozhukov et al. \(2014\)](#)) applied to the linear expansion of the bias-corrected estimator. This approximation is justified by the fact that, after bias correction, the stochastic fluctuation $\widehat{\eta}^*(\mathbf{x}) - \eta(\mathbf{x})$ can be expressed asymptotically as a weighted sum of independent errors as

$$\widehat{Z}_n(\mathbf{x}) \approx \sum_{i=1}^n a_i(\mathbf{x}) \cdot \varepsilon_i, \quad (6)$$

where $a_i(\mathbf{x})$ are data-dependent kernel weights that are fixed conditional on the stage-1 empirical data $\mathcal{D}_1$, and ε_i are the actual randomness driving the estimator. In the present setting,

$$a_i(\mathbf{x}) = \frac{1}{\widehat{\sigma}(\mathbf{x})} \left(\mathbf{1}_{\{T_i=1\}} \cdot w_{1,i}(\mathbf{x}) \cdot \widehat{\varepsilon}_{1,i} - \mathbf{1}_{\{T_i=0\}} \cdot w_{0,i}(\mathbf{x}) \cdot \widehat{\varepsilon}_{0,i} \right),$$

where $w_{t,i}(\mathbf{x})$ denote local linear kernel weights, $\widehat{\varepsilon}_{t,i}$ are mean-zero residuals associated with treatment group $t \in \{0, 1\}$, and $\widehat{\sigma}(\mathbf{x})$ is assumed to be strictly positive on $\mathcal{X}$. The multiplier bootstrap replaces the unknown errors ε_i with i.i.d. standard normal variables

$$\xi_i \sim \mathcal{N}(0, 1),$$

following the standard Gaussian multiplier bootstrap construction in [Armstrong and Shen \(2023\)](#) and [Chernozhukov et al. \(2014\)](#), yielding the bootstrap process

$$\widehat{Z}_n^*(\mathbf{x}) = \sum_{i=1}^n a_i(\mathbf{x}) \cdot \xi_i.$$

Conditional on $\mathcal{D}_1$, the empirical distribution of $\sup_{\mathbf{x} \in \mathcal{X}} \widehat{Z}_n^*(\mathbf{x})$ provides a valid approximation to the distribution of $\sup_{\mathbf{x} \in \mathcal{X}} \widehat{Z}_n(\mathbf{x})$, and the critical value $C_{1-\alpha}$ is estimated as the corresponding $(1 - \alpha)^{\text{th}}$ quantile of the bootstrap distribution.

Using this critical value $C_{1-\alpha}$, we construct a one-sided confidence set for the treatment-effect level set. Specifically, for a given threshold τ , the estimated subgroup is defined as

$$\widehat{L}_\tau = \{\mathbf{x} \in \mathcal{X} : \widehat{\eta}^*(\mathbf{x}) \geq \tau - C_{1-\alpha} \widehat{\sigma}(\mathbf{x})\}. \quad (7)$$

By construction, this set satisfies

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(L_\tau \subseteq \widehat{L}_\tau \right) \geq 1 - \alpha, \quad (8)$$

providing a one-sided confidence guarantee that prioritizes protection against false exclusion of truly benefitting patients. The resulting estimator yields a smooth, interpretable subgroup suitable for adaptive enrollment in the confirmatory stage.

Proposition 1 *Let $\eta(\mathbf{x})$ denote the smooth conditional treatment effect function defined on $\mathcal{X} = \text{supp}(X) \subset \mathbb{R}^d$. Assume that $\eta(\mathbf{x})$ is three times continuously differentiable so that the second-order bias expansion for local linear estimators and the explicit bias correction procedure motivated by [Qiao and Polonik \(2019\)](#) are valid. Set*

$$L_\tau = \{\mathbf{x} \in \mathcal{X} : \eta(\mathbf{x}) \geq \tau\}$$

be the corresponding treatment-effect level set for a fixed threshold τ . Suppose $\widehat{\eta}^(\mathbf{x})$ is a bias corrected local linear estimator of $\eta(\mathbf{x})$ with pointwise standard deviation estimator $\widehat{\sigma}(\mathbf{x})$. Define*

$$\widehat{L}_\tau = \{\mathbf{x} \in \mathcal{X} : \widehat{\eta}^*(\mathbf{x}) \geq \tau - C_{1-\alpha} \widehat{\sigma}(\mathbf{x})\},$$

where $C_{1-\alpha}$ is a critical value satisfying equation (5). Then the estimated level set $\widehat{L}_\tau$ satisfies the one-sided coverage property

$$\liminf_{n \rightarrow \infty} \mathbb{P} \left(L_\tau \subseteq \widehat{L}_\tau \right) \geq 1 - \alpha.$$

The proof strategy combines one-sided confidence set construction for regression level sets in [Wan et al. \(2024\)](#) with the Gaussian multiplier bootstrap framework of [Armstrong and Shen \(2023\)](#) and the explicit bias-correction approach of [Qiao and Polonik \(2019\)](#).

Proof Let $g(\mathbf{x}) = -\eta(\mathbf{x})$ and according to (2), we have

$$\mathcal{S}_\tau = \{\mathbf{x} \in \mathcal{X} : g(\mathbf{x}) > -\tau\},$$

Let $\widehat{g}_{bc}(\mathbf{x}) = \widehat{g}(\mathbf{x}) - \widehat{\beta}$ denotes a bias-corrected estimator of $\widehat{g}(\mathbf{x})$ where $\widehat{\beta} = \mathbb{E}(\widehat{g}(\mathbf{x})) - g(\mathbf{x})$, and $\widehat{\sigma}(\mathbf{x})$ be the standard deviation of $\widehat{g}_{bc}(\mathbf{x})$. Then (4) is updated to be

$$\widehat{Z}_n(\mathbf{x}) = \frac{\widehat{g}^*(\mathbf{x}) - g(\mathbf{x})}{\widehat{\sigma}(\mathbf{x})} \quad \mathbf{x} \in \mathcal{X},$$

and $\{\mathbf{x} \in \mathcal{S}_\tau\} = \{\forall \mathbf{x} \in \mathcal{X} : g(\mathbf{x}) > -\tau\}$. Then

$$\begin{aligned} \left\{ \sup_{\mathbf{x} \in \mathcal{X}} \widehat{Z}_n(\mathbf{x}) \leq C_{1-\alpha} \right\} &= \left\{ \forall \mathbf{x} \in \mathcal{X}, \frac{\widehat{g}^*(\mathbf{x}) - g(\mathbf{x})}{\widehat{\sigma}(\mathbf{x})} \leq C_{1-\alpha} \right\} \\ &= \left\{ \forall \mathbf{x} \in \mathcal{X}, g(\mathbf{x}) \geq \widehat{g}^*(\mathbf{x}) - \widehat{\sigma}(\mathbf{x}) \cdot C_{1-\alpha} \right\}. \end{aligned}$$

Define $\widehat{\mathcal{S}}_\tau = \{\mathbf{x} \in \mathcal{X} : \widehat{g}^*(\mathbf{x}) \geq -\tau + \widehat{\sigma}(\mathbf{x}) \cdot C_{1-\alpha}\}$, and for any $\mathbf{x} \in \widehat{\mathcal{S}}_\tau$,

$$\begin{aligned} \left\{ \sup_{\mathbf{x} \in \mathcal{X}} \widehat{Z}_n(\mathbf{x}) \leq C_{1-\alpha} \right\} &= \left\{ \forall \mathbf{x} \in \mathcal{X} : \text{s.t. } \mathbf{x} \in \widehat{\mathcal{S}}_\tau, \widehat{g}^*(\mathbf{x}) \leq g(\mathbf{x}) + \widehat{\sigma}(\mathbf{x}) \cdot C_{1-\alpha} \right\} \\ &= \left\{ \forall \mathbf{x} \in \widehat{\mathcal{S}}_\tau : -\tau + \widehat{\sigma}(\mathbf{x}) \cdot C_{1-\alpha} \leq \widehat{g}^*(\mathbf{x}) \leq g(\mathbf{x}) + \widehat{\sigma}(\mathbf{x}) \cdot C_{1-\alpha} \right\} \\ &= \left\{ \forall \mathbf{x} \in \widehat{\mathcal{S}}_\tau : -\tau \leq \widehat{g}^*(\mathbf{x}) \leq g(\mathbf{x}) \right\} \end{aligned}$$

$$\begin{aligned}
&\subseteq \{\forall \mathbf{x} \in \widehat{\mathcal{S}}_\tau : -\tau \leq g(\mathbf{x})\} \\
&= \{\widehat{\mathcal{S}}_\tau \subseteq \mathcal{S}_\tau\}.
\end{aligned}$$

Hence,

$$\{\sup_{\mathbf{x} \in \mathcal{X}} \widehat{Z}_n(\mathbf{x}) \leq C_{1-\alpha}\} \subseteq \{\widehat{\mathcal{S}}_\tau \subseteq \mathcal{S}_\tau\}.$$

Accordingly, $\mathbb{P}(\sup_{\mathbf{x} \in \mathcal{X}} \widehat{Z}_n(\mathbf{x}) \leq C_{1-\alpha}) \leq \mathbb{P}(\widehat{\mathcal{S}}_\tau \subseteq \mathcal{S}_\tau)$ by monotonicity, and then

$$1 - \alpha \leq \liminf_n \mathbb{P}(\sup_{\mathbf{x} \in \mathcal{X}} \widehat{Z}_n(\mathbf{x}) \leq C_{1-\alpha}) \leq \liminf_n \mathbb{P}(\widehat{\mathcal{S}}_\tau \subseteq \mathcal{S}_\tau).$$

Finally, since $\widehat{L}_\tau = \widehat{\mathcal{S}}_\tau^c$, and $\widehat{\eta}^*(\mathbf{x}) = -\widehat{g}^*(\mathbf{x})$, we obtain (8) as the desired one-sided confidence guarantee for level set $\widehat{L}_\tau$,

$$\widehat{L}_\tau = \{\mathbf{x} \in \mathcal{X} : \widehat{\eta}^*(\mathbf{x}) > \tau - C_{1-\alpha} \cdot \widehat{\sigma}(\mathbf{x})\}.$$

□

4 Inference

A fundamental challenge arises because subgroup membership is determined using stage 1 data, inducing dependence between subgroup selection and subsequent inference. As a result, unadjusted analyses may inflate the type I error rate due to reuse of the same data for subgroup selection and confirmatory testing, a phenomenon commonly referred to as double dipping (Cox (1975)). Preserving valid inference therefore requires explicit adjustment for the adaptive subgroup selection mechanism, as well as ensuring an appropriate type I error rate control (U.S. Food and Drug Administration (2019)). We now discuss four techniques that have been proposed to control type I error rate in two-stage adaptive designs.

Wassmer and Brannath (2016) proposed a closed testing framework for adaptive clinical trials that combines stage-wise combination tests and conditional error arguments to preserve strong family-wise type I error control under data-driven design modifications. In adaptive enrichment settings, this approach provides an inferential strategy when multiple candidate subgroups or population hypotheses are considered simultaneously. Valid confirmatory inference is obtained by testing all intersection hypotheses associated with the candidate subgroup family, and an elementary hypothesis is rejected only if all corresponding intersection hypotheses are also rejected. The resulting procedure naturally accommodates hierarchical or nested subgroup structures and is particularly well suited for subgroup selection based on continuous biomarkers.

When subgroup selection is based on a continuous biomarker, the collection of candidate subgroups can be viewed as a nested family indexed by threshold values (monotonicity of the subgroups is assumed), as discussed in Stallard (2023). In this case, treatment effect estimates across subgroups exhibit a dependence structure analogous to that arising in group-sequential settings. This connection enables the construction of valid tests by combining stage-wise evidence, where the stage 1 component is adjusted to reflect the data-driven selection of the subgroup and then combined with independent stage 2 information. Depending on the number of candidate thresholds, the resulting inference can be formulated using multivariate normal

approximations or Brownian motion limits, providing a practical way to control the family-wise type I error rate while retaining flexibility in the subgroup selection rule.

In the two-stage adaptive framework, selective randomization-based inference (Freidling et al. (2024)) is a model-agnostic approach. Because treatment assignments are randomized within each stage, valid tests can be constructed by conditioning on the randomization mechanism and the information used in subgroup selection. Selective randomization tests, in particular, provide a natural way to evaluate treatment effects while mitigating the impact of data-adaptive enrichment.

A likelihood-based inferential framework for two-stage enrichment designs with continuous biomarkers can be built by explicitly modeling the joint distribution of outcomes and biomarkers, as in Lin et al. (2021). In this setting, the enrichment threshold selected at the interim analysis introduces dependence between stage 1 and stage 2 samples, which leads to nonstandard asymptotic behavior. Under a bivariate normal formulation, the resulting maximum likelihood estimators follow a randomly scaled mixture of normal distributions rather than classical normal limits. By using random information measures and stable convergence arguments, valid test statistics can be constructed for both the overall treatment effect and the association between outcome and biomarker. This formulation allows inference to proceed using data from both stages while properly accounting for the randomness of the selected subgroup.

Taken together, these inferential strategies within the two-stage adaptive enrichment setting are designed to address the same fundamental issue: preserving valid inference under data-driven subgroup selection while maintaining appropriate control of the type I error rate. Within our proposed enrichment framework, we aim to further investigate and compare these approaches to better understand their relative performance in terms of error control and efficiency (work in progress).

Power in the confirmatory stage is improved primarily through adaptive enrichment, which concentrates enrollment on treatment-sensitive patients. This gain is further supported by one-sided confidence constructions that preserve subgroup size and by inference strategies that allow efficient reuse of learning-stage data under valid error control.

Beyond confirmatory testing, another objective of this stage is to re-estimate the benefitting subgroup, particularly the threshold, using data from both stages. Although this step is not part of clinical decision making, it can provide valuable insight for subsequent treatment development by refining understanding of the disease mechanism and identifying patient populations most likely to benefit from the therapy.

The development and comparison of confirmatory inference procedures for this setting, including their finite-sample properties and computational feasibility, remain important topics for ongoing research. The present work establishes a coherent link between nonparametric subgroup identification and valid confirmatory analysis in two-stage adaptive enrichment designs.

5 Simulation Study

A simulation study is conducted to examine the finite-sample performance of the proposed regression level-set estimator in the learning stage. The primary goal is

to assess one-sided coverage of the true benefitting subgroup and to evaluate false exclusion behavior under moderate sample sizes typical of clinical trials.

5.1 Design

We consider a univariate biomarker setting for clarity of presentation. The biomarker is generated as $X \sim \text{U}(0, 1)$, and an independent random component is generated as $U \sim \text{U}(0, 1)$. As in [Armstrong and Shen \(2023\)](#), the baseline and treated mean response functions are specified as

$$\mu_0(x) = 5(x - 0.5)^2, \quad \mu_1(x) = \mu_0(x) + 5(x - 0.5)^2,$$

yielding a smooth, non-monotone conditional treatment effect function. Treatment assignment follows a covariate-dependent randomized rule $T = 1\{0.1X + U > 0.55\}$, and outcomes are generated as

$$Y = \mu_0(X) + T \cdot \mu_1(X) + \varepsilon,$$

where $\varepsilon \sim \mathcal{N}(0, 1/9)$. The benefitting subgroup is defined through treatment-effect level sets $L_\tau = \{x : \eta(x) \geq \tau\}$, with threshold values τ ranging from cases where the entire population benefits to cases where no benefitting subgroup exists. Sample sizes vary from $n = 100$ to $n = 600$, reflecting enrollment sizes commonly encountered in enrichment trials.

Treatment effects are estimated using local linear regression with explicit bias correction with the Gaussian kernel function adopted as

$$K(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right), \quad u \in \mathbb{R}.$$

Critical values for one-sided confidence sets are obtained via a Gaussian multiplier bootstrap approximation to the supremum of the studentized error process as in [\(5\)](#).

5.2 Coverage Probabilities

Across a wide range of thresholds and sample sizes, the proposed estimator achieves high empirical coverage of the true treatment-effect level set. Figures [\(1,2\)](#) are generated using $n = 200$ observations, illustrate the qualitative behavior of the estimated level sets across representative threshold values, showing that the estimator adapts smoothly as the definition of benefit varies.

In each plot, the black curve represents the true treatment effect function $\eta(x)$, while the blue curve denotes its estimated counterpart $\hat{\eta}(x)$. The horizontal line segments in green and red correspond to the estimated and true level sets. It is important to note that the vertical position of these segments does not carry substantive meaning; rather, their horizontal extent reflects the length of the corresponding intervals, indicating the regions identified as belonging to the level set. For Figure [2](#) and the bottom part in Figure [1](#), the gray dashed horizontal line indicates the threshold level τ on

the treatment effect scale, and the point markers indicate the status of the estimated treatment effect $\hat{\eta}(x)$ relative to the threshold τ , where the solid circles denote points with $\hat{\eta}(x) \geq \tau$ (upper region) and crosses denote points with $\hat{\eta}(x) < \tau$ (lower region).

When $\tau = 0$ in Figure 1, the true benefitting subgroup coincides with the full population, and the estimated level set correctly avoids excluding patients. As τ increases (e.g., $\tau = 0.6$) Figure 1, the benefitting subgroup becomes more localized, and the estimated level set contracts accordingly in a smooth and stable manner, closely tracking the true boundary without excessive variability. This demonstrates the estimator’s ability to adapt to intermediate regimes where only a subset of patients benefit from treatment.

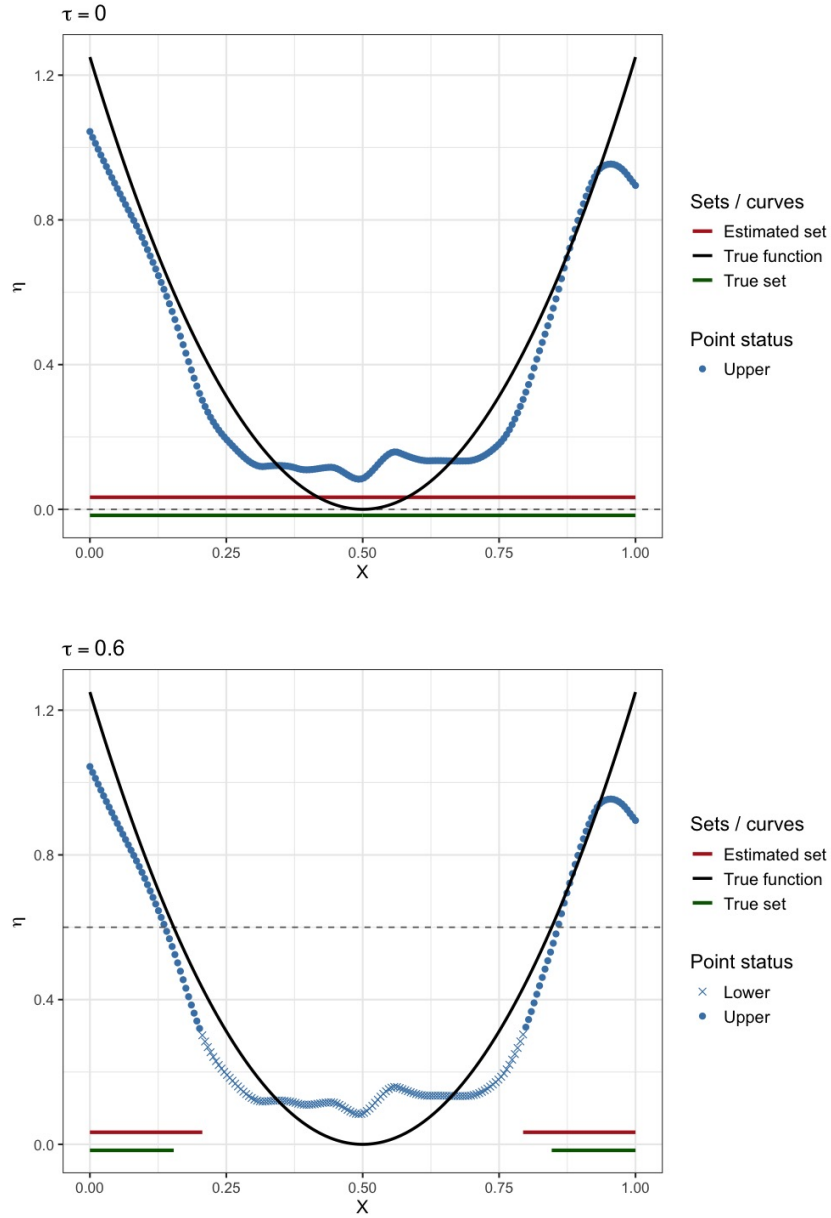


Fig. 1: Estimated and true regression level sets for two threshold values. Top: $\tau = 0$, where the full population is the benefiting group. Bottom: $\tau = 0.6$, where a subpopulation is the benefiting group.

In the extreme case where no benefiting subgroup exists (e.g., $\tau = 1.4$), the estimator appropriately yields an empty set, avoiding spurious inclusion of patients.

Across all scenarios, the estimated level sets align well with the true underlying structure, illustrating both accurate boundary recovery and robustness to changes in the threshold parameter.

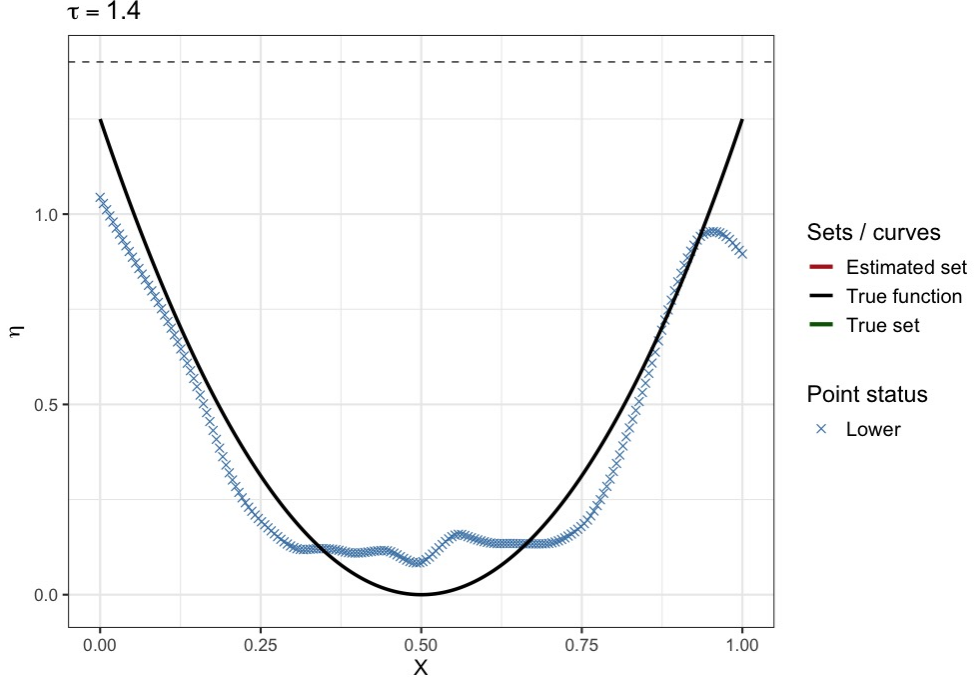


Fig. 2: Estimated and true regression level sets for threshold value $\tau = 1.4$, and the benefitting subgroup does not exist.

Table (1) shows that the proposed level-set estimator achieves consistently high empirical coverage for all considered enrollment sizes, with the exception of the smallest sample size ($n = 100$). For enrollment sizes of 200 or larger, the observed coverage probabilities are close to or exceed the nominal level $1 - \alpha$ specified in (1), indicating that the method does not exhibit systematic under-coverage in finite samples and provides reliable protection against erroneous subgroup exclusion. When the enrollment size is limited to 100 patients, the empirical coverage probability falls below the nominal level. This behavior is likely attributable to data sparsity within treatment arms, which increases estimation variability and degrades finite-sample performance in small samples.

5.3 Threshold Comparison

To illustrate the empirical behavior of the threshold specifications introduced in Section 2.3, we compare the fixed clinically meaningful threshold described in

Table 1: Coverage probabilities for different enrollment sizes

Number of enrollment	Coverage probability
100	0.935
200	0.990
300	0.993
400	0.994
500	0.998
600	0.997

Empirical coverage probabilities are computed using 1000 iterations for each enrollment size.

Section 2.3 (i) with the average-effect calibrated threshold described in Section 2.3 (ii) under the same simulation setting. For simplicity, we restrict attention to these two specifications, reflecting the distinction between a fixed clinically chosen threshold and a data-adaptive threshold estimated from the population treatment effects. However, the one-sided coverage probabilities reported in Table 1 are primarily determined by the uniform confidence construction for the treatment-effect function rather than by the particular choice of threshold. Once a threshold is specified or estimated and plugged into the level-set definition as in (2), the confidence set is obtained using the same confidence-band construction using (7). Consequently, the threshold primarily affects the estimated enrollment region and the number of eligible patients available for enrollment, while the underlying coverage mechanism remains unchanged.

As Figure 3 illustrates, the average-effect calibrated threshold $\tau^* = 0.16$ is obtained using the average-effect calibrated threshold definition described in Section 2.3(ii), with the same level corresponding to the fixed threshold $\tau = 0.6$.

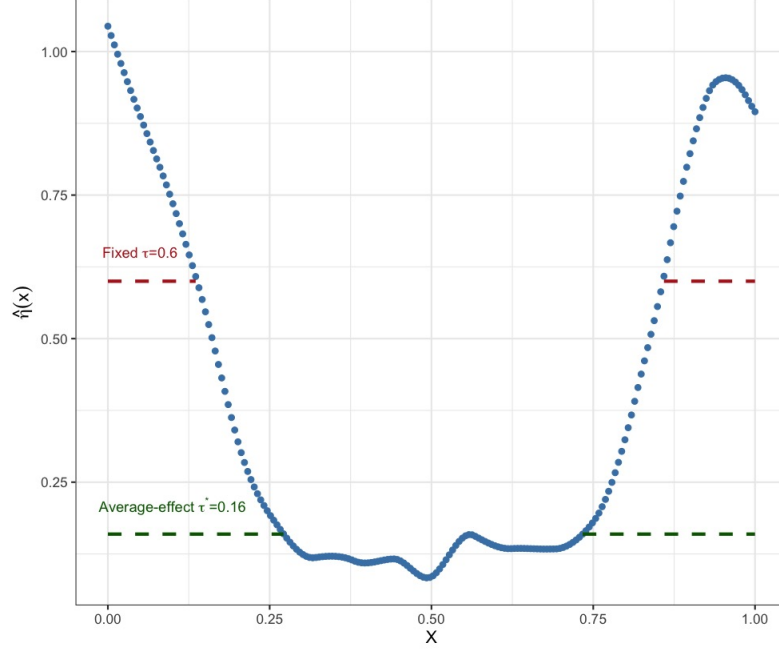


Fig. 3: Comparison of fixed clinically meaningful and average-effect calibrated threshold specifications. Horizontal segments indicate the estimated subgroup regions satisfying $\hat{\eta}(x) \geq \tau$ under each threshold definition.

6 Discussion

We develop a two-stage adaptive enrichment framework based on nonparametric regression level-set estimation. In the learning stage, bias-corrected kernel estimators are used to identify benefitting subgroups through a one-sided confidence construction that controls false exclusion with high probability. The resulting subgroup estimates are smooth, interpretable, and naturally suited for adaptive enrollment in a confirmatory stage, while accommodating nonlinear and non-monotone biomarker–treatment relationships.

From a design perspective, emphasizing protection against false exclusion aligns statistical inference with ethical considerations in patient enrollment by reducing the risk of denying potentially beneficial treatment. Extending the framework to multiple biomarkers remains future work and would require the use of dimensionality reduction techniques, such as Lasso. Ongoing work focuses on refining confirmatory inference, that will ensure type I error rate control. Future work involves comparative inference to determine the most powerful method to test our hypotheses of interest.

7 Statements and Declarations

The authors thank the expert reviewers for their thoughtful comments and declare that they have no competing interests.

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